

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

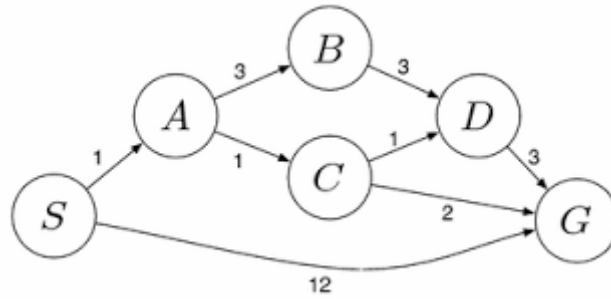
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ARTIFICIAL INTELLIGENCE (CSA1703)

Assessment Tool – 1

CO1: Intelligent Agents and Uninformed Search Methods

Question 1: Water Jug Problem

Problem

Two jugs of capacities **4 gallons** and **3 gallons** are given. Obtain exactly **2 gallons** in the 4-gallon jug.

Solution

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour into the 4-gallon jug until it becomes full	4	2
5	Empty the 4-gallon jug	0	2
6	Transfer the remaining 2 gallons into the 4-gallon jug	2	0

Result

Hence, exactly **2 gallons** of water is obtained in the **4-gallon jug**.

Question 2: Mars Rover Intelligent Agent

i) Percepts

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Chemical sensor readings
- Terrain information

ii) Operating Environment

- Partially Observable
- Deterministic
- Sequential

- Dynamic
- Continuous
- Single Agent

iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Send collected data to Earth

iv) Performance Measure

- Accuracy of scientific data
- Number of samples collected
- Safe navigation
- Efficient energy usage
- Successful mission completion

v) Suitable Agent Architecture

Model-Based Goal-Based Agent

Reason:

The Mars Rover maintains knowledge of its environment, plans its path, avoids obstacles, and achieves mission goals autonomously.

Question 3: 8-Queens Problem

Problem Formulation

Initial State

- Empty chessboard.

State Space

- All possible arrangements of eight queens on the chessboard.

Operators

- Place one queen in a safe position.

Goal State

- No two queens attack each other horizontally, vertically, or diagonally.

Path Cost

- Each queen placement has a cost of one.

Search Strategy

- Backtracking using Depth-First Search (DFS).
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Question 4: OLA Cab Booking**Type of Problem-Solving Agent****Goal-Based Agent****Reason**

The customer's goal is to reach the destination by selecting the most suitable cab based on comfort, availability, and fare.

Algorithm (Pseudo Code)

START

Input Pickup Location

Input Destination

Input Preferred Cab Type

Display Available Cabs

IF Preferred Cab is Available THEN

 Select Preferred Cab

ELSE

 Select Nearest Available Cab

ENDIF

Calculate Fare

Confirm Booking

Assign Driver

Display Driver Details

STOP

Question 5: Uniform Cost Search (UCS)**Given Graph**

Path Cost

S → A 1

Path Cost

$S \rightarrow G$ 12

$A \rightarrow B$ 3

$A \rightarrow C$ 1

$B \rightarrow D$ 3

$C \rightarrow D$ 1

$C \rightarrow G$ 2

$D \rightarrow G$ 3

UCS Expansion

Expanded Path	Total Cost
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S	0
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$S \rightarrow A$	1
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$S \rightarrow A \rightarrow C$	2
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$S \rightarrow A \rightarrow C \rightarrow G$	4
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$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6
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$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10
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$S \rightarrow G$	12
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Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

4

Conclusion

Uniform Cost Search expands the node with the minimum cumulative cost and guarantees the optimal solution when all path costs are non-negative.

This format is suitable for writing directly as an **engineering assignment**, with proper headings, subheadings, tables, and conclusions.